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Inflation Expectations and Firm Decisions: New Causal Evidence

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1 Big picture

- **Question.** Do firms' inflation expectations causally affect their real and financial decisions, including prices, employment, investment, credit demand, balance-sheet composition, and profits?
- **Why the question matters.** Modern macroeconomic policy increasingly relies on expectations management, especially when nominal interest rates are close to the effective lower bound (ELB). If firms do not react to inflation expectations, forward guidance and other communication policies have weaker real effects.
- **Core empirical problem.** Inflation expectations are endogenous. Firms expecting higher inflation may also be observing different demand, cost, or credit conditions. A simple correlation between inflation expectations and firm behavior cannot be interpreted causally.
- **Identification strategy.** The paper exploits a randomized information treatment in the Bank of Italy's Survey on Inflation and Growth Expectations (SIGE). A randomly selected treatment group repeatedly receives recent public information about inflation before reporting inflation expectations; a control group does not.
- **First-stage result.** The treatment moves inflation expectations strongly. A 1 percentage point higher inflation signal raises six-month-ahead expectations by 0.590 percentage points and one-year-ahead expectations by 0.548 percentage points.
- **Main full-sample result.** Higher inflation expectations lead firms to raise prices only slightly and temporarily, but to reduce employment, reduce fixed assets, reduce sales, increase credit utilization, increase loan applications, raise inventories, and raise leverage.
- **Mechanism outside the ELB.** Italian firms interpret higher inflation as bad news, similar to a supply-side or stagflationary signal. They expect worse aggregate conditions, lower product demand, worse liquidity, and weaker access to credit.
- **ELB result.** During the ELB period, higher inflation expectations generate stronger price increases and no employment decline. Firms instead associate higher inflation with better demand and improved business conditions.
- **Profit result.** Treated firms earn modestly higher profit shares during the treatment period, while no difference exists before treatment. This is consistent with rational inattention: the information is useful, but the private payoff from tracking inflation is small.
- **Economic takeaway.** Inflation expectations are not merely survey answers. They causally change firm behavior, but the direction of the effect depends on the macroeconomic environment and on how firms interpret inflation news.

2 Incremental contribution of the paper

- **Firm expectations rather than household expectations.** Much of the inflation-expectations literature studies households. This paper focuses on firms, whose decisions directly shape prices, employment, investment, production, and credit demand.
- **Causal evidence from randomized information.** The paper uses randomized provision of inflation information as an instrument for inflation expectations. This addresses the endogeneity problem that limits many studies of expectations and behavior.

- **Repeated treatment over many years.** Unlike one-shot experiments, the Italian survey repeatedly treats firms across more than six years. This creates persistent and time-varying first-stage variation in inflation expectations.
- **Rich administrative data linkage.** The survey is linked to the Italian Central Credit Registry, INPS social-security records, and CADS balance-sheet data. This lets the authors study actual employment, credit, investment, leverage, sales, inventories, wages, and profits rather than only survey intentions.
- **Dynamic response horizons.** The paper traces how effects evolve across several quarters and annual horizons. This matters because prices, hiring, investment, and balance-sheet choices adjust at different speeds.
- **ELB comparison.** The paper compares behavior inside and outside the effective lower bound period. This directly tests a central New Keynesian prediction: higher expected inflation is more expansionary when nominal rates do not rise.
- **Rational inattention evidence.** The treatment affects firms' profits positively but only modestly. This explains why firms may rationally choose not to track inflation closely even though free information changes their decisions.
- **Context dependence of inflation interpretation.** The paper shows that the same inflation signal can be interpreted differently across macroeconomic regimes. Outside the ELB, higher inflation is read as bad news; at the ELB, it is read as stronger demand.

3 Data and measurement

3.1 Survey on Inflation and Growth Expectations (SIGE)

SIGE is a quarterly business survey run by the Bank of Italy since 1999. It covers firms with at least 50 workers operating in industry, nonfinancial private services, and, from 2013 onward, construction. Recent waves contain about 1,000 firms. The survey asks about inflation expectations, expected general economic conditions, firm-specific business conditions, demand, employment, investment, liquidity, credit access, price changes, and the reasons behind future price changes.

The survey is central because it provides both expectations and the randomized information treatment. It also gives some direct decisions such as reported employment and price changes.

3.2 Information treatment

Before answering inflation-expectations questions, treated firms receive recent public information about the most recent annual inflation rate in Italy and the euro area. Control firms are asked for their inflation expectations without this information. The treatment is randomized across firms.

For firm i in survey wave t , the treatment variable is

$$T_t^i = \begin{cases} \text{recent inflation rate,} & \text{if firm } i \text{ is treated in wave } t, \\ 0, & \text{if firm } i \text{ is in the control group.} \end{cases}$$

Because actual inflation changes over time, the treatment is not simply a binary indicator. It varies both across firms and across survey waves.

3.3 Italian Central Credit Registry (CCR)

The CCR contains detailed information on credit granted by banks and financial intermediaries operating in Italy. The paper uses two main credit outcomes:

- **Credit-line utilization:** the ratio of credit drawn to credit available on overdraft credit lines.
- **Loan applications:** requests by banks to obtain preliminary information about potential borrowers, interpreted as applications to new lenders.

Credit utilization is useful because it is closer to credit demand by the firm and less contaminated by supply-side changes than new committed credit.

3.4 INPS employment and wage records

INPS provides monthly administrative records on employment, wages, occupational category, and contract type. The paper uses these data to validate survey-reported employment and to study composition effects across white-collar versus blue-collar workers and temporary versus permanent workers.

3.5 CADS balance-sheet and income-statement data

CADS contains financial statements for Italian limited liability firms. The paper uses CADS to study fixed assets, sales, liquidity, trade credit, cash, inventories, leverage, and profits. These variables are available annually rather than quarterly.

4 Identification and empirical specifications

4.1 First stage: treatment and inflation expectations

The first-stage regression measures how the information treatment changes inflation expectations:

$$F_t^i \pi(h) = \alpha_h + \beta_h T_t^i + error_{t,h}^i.$$

Here $F_t^i \pi(h)$ is firm i 's inflation expectation at horizon h in wave t . The coefficient β_h captures how strongly firms update expectations when given recent inflation information.

The treatment is strong at all horizons but weaker at longer horizons. This means firms' long-run inflation expectations are harder to move than short-run expectations.

4.2 Dynamic persistence of the information treatment

To study whether information has persistent effects on expectations, the paper estimates

$$F_t^i \pi(h) = \alpha_h + \beta_{h,0} T_t^i + \beta_{h,1} T_{t-1}^i + \beta_{h,2} T_{t-2}^i + \cdots + \beta_{h,q} T_{t-q}^i + error_{t,h}^i.$$

The main finding is that the contemporaneous treatment matters strongly, the previous-quarter treatment has only limited effects, and older treatments have little independent effect. The implication is that inflation information must be communicated repeatedly if policymakers want to create persistent changes in beliefs.

4.3 IV specification for firm decisions

To estimate the causal effect of inflation expectations on decisions, the paper estimates

$$y_{t+k}^i = \alpha_k + \gamma_k F_{t-1}^i \pi(12m) + controls_{t-2}^i + error_{t-1,t+k}^i,$$

where $F_{t-1}^i \pi(12m)$ is one-year-ahead inflation expectations and is instrumented using the information treatment T_{t-1}^i .

The controls include firms' lagged expectations about business conditions, employment growth, liquidity, current general economic conditions, and the probability of improvement in Italy's general economic situation.

The key exclusion restriction is that the information treatment affects firm decisions only through inflation expectations and the other expectations controlled for.

4.4 Annual specifications

For annual variables, the paper uses inflation expectations from the fourth quarter of year $t - 1$ and measures outcomes as changes from year $t - 1$ to year $t + 1$. This is used for fixed assets, sales, balance-sheet ratios, wages, employment composition, and profits.

4.5 ELB subsample

The effective lower bound sample runs from 2014Q4 to 2018Q2. The paper repeats the IV specifications in this period to test whether the effects of inflation expectations change when nominal policy rates are constrained.

5 Tables and figures

5.1 Figures I and II: Treatment, inflation expectations, and forecast distributions

Read:

- Figure I shows the time series of inflation expectations for treated and control firms. Before the differential treatment, the two groups track each other closely. After 2012Q3, the treated group follows actual inflation much more closely.
- Panel A shows that when inflation falls from 2012 to 2015, treated firms' expectations fall faster than control firms' expectations. When inflation rises in 2017, treated firms' expectations rise faster.

- Panel B shows that the treatment reduces disagreement: the control group has systematically more dispersed inflation expectations after treatment begins.
- Figure II plots kernel densities of one-year-ahead inflation expectations in selected waves. The control distribution is more dispersed, while the treated distribution is more concentrated around the recent inflation signal.

Economic message: The treatment creates large, visible, and repeated differences in beliefs. It gives the paper a powerful first stage and also shows that firms are not fully attentive to publicly available inflation information.

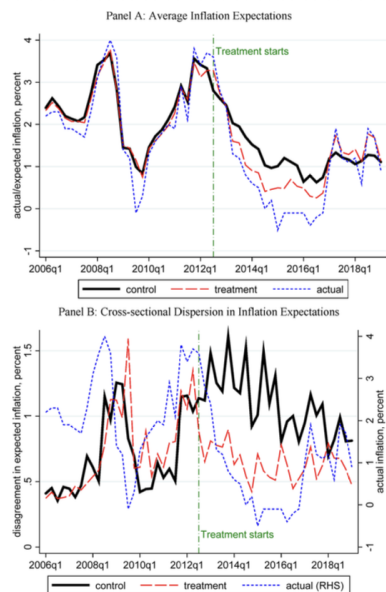


FIGURE I

Time Series of Inflation Expectations for Treatment and Control Groups

Treated firms are presented with the most recent value of actual inflation, which is shown with blue, short-dash line (color version of figure available online). We use treatment assignment in 2012Q4 to classify firms into treatment and control groups for the period 2006Q1–2012Q2.

(a) Figure I: Time series of inflation expectations for treatment and control firms.

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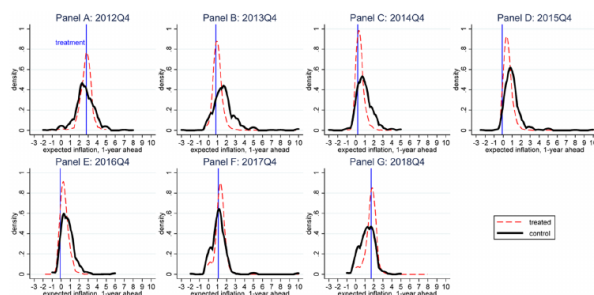


FIGURE II

Distribution of Inflation Expectations for Treated and Control Firms

Each panel plots the kernel density of inflation expectations (one year ahead) for treated and control firms in specific survey waves indicated in the title of each panel. The bandwidth is 0.9. The vertical thin blue line shows the inflation rate.

(b) Figure II: Distribution of inflation expectations in selected waves.

5.2 Figures III and IV: Horizons and price-setting channels

Read:

- Figure III shows the distribution of expectations by horizon in 2014Q4. The treatment shifts expectations at six months, one year, two years, and four years, although the shift is weaker at longer horizons.
- This visual result matches Table I: short-run expectations respond most, and long-run expectations respond less.
- Figure IV shows the factors that firms report as contributing to expected price changes: total demand, raw material prices, labor costs, and competitors' prices.
- Raw material prices are an important upward price force, while demand and competitor pricing

often exert downward pressure.

Economic message: Firms interpret inflation news through multiple channels. Outside the ELB, higher inflation expectations mainly signal higher input costs and weaker demand, rather than a pure demand boom.

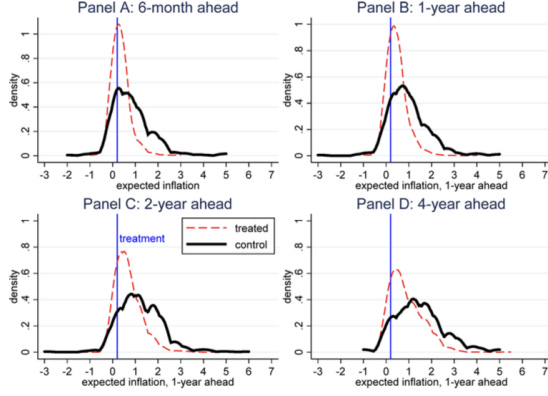


FIGURE III

Distribution of Inflation Expectations by Horizon for Treated and Control Firms, 2014Q4

Each panel plots the kernel density of inflation expectations by forecast horizon (indicated in panel titles) for treated and control firms in the 2014Q4 wave of the survey. The bandwidth is 0.2. The vertical, thin blue line shows the inflation rate given to treated firms.

(a) Figure III: Expectations by horizon, treated versus control firms.

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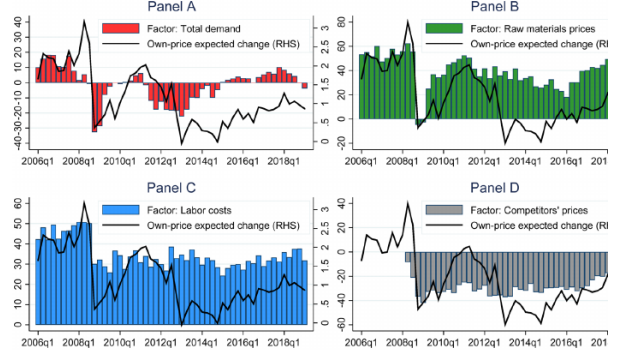


FIGURE IV

Underlying Factors Contributing to Expected Price Changes

(b) Figure IV: Underlying factors behind expected price changes.

5.3 Tables I and II: First stage and persistence of signals

Read:

- Table I estimates the direct effect of the information treatment on inflation expectations by horizon.
- A 1 percentage point higher inflation signal raises six-month-ahead expectations by 0.590 percentage points, one-year-ahead expectations by 0.548 percentage points, two-year-ahead expectations by 0.471 percentage points, and four-year-ahead expectations by 0.369 percentage points.
- The first stage is statistically strong at every horizon, but the R^2 declines with horizon, meaning long-run beliefs are harder to move.
- Table II studies duration. The contemporaneous treatment remains large and significant. The previous-quarter signal has at most a small effect, and older signals have little stable explanatory power.

Economic message: Firms update strongly when information is supplied, but the effect fades quickly. Expectations policy therefore requires repeated and persistent communication, not one-time announcements.

TABLE II
DURATION OF EFFECTS OF SIGNALS ON INFLATION EXPECTATIONS

	$F_{t-1}^i \pi^{(12m)}$ (1)	$F_t^i \pi^{(12m)}$ (2)	$F_{t-1}^i \pi^{(12m)}$ (3)	$F_t^i \pi^{(12m)}$ (4)	$F_{t-1}^i \pi^{(12m)}$ (5)
T_t^i	0.548*** (0.057)	0.381*** (0.049)	0.443*** (0.059)	0.435*** (0.059)	0.465*** (0.064)
T_{t-1}^i		0.152* (0.074)	-0.044 (0.077)	0.083 (0.059)	0.069 (0.075)
T_{t-2}^i			0.104 (0.085)	-0.135** (0.060)	-0.067 (0.051)
T_{t-3}^i				0.103 (0.066)	-0.053 (0.056)
T_{t-4}^i					0.056 (0.076)

TABLE I
EFFECT OF THE TREATMENT WITH PAST INFLATION ON INFLATION EXPECTATIONS

Dependent variable: Inflation expectations by horizon, $F_t^i \pi^{(h)}$

	6 months ahead (1)	1 year ahead (2)	2 years ahead (3)	4 years ahead (4)
T_t^i	0.590*** (0.060)	0.548*** (0.057)	0.471*** (0.050)	0.369*** (0.046)
Observations	25,531	25,531	25,531	20,007
R-squared	0.237	0.204	0.146	0.053
Sample	2012Q3–2019Q1	2012Q3–2019Q1	2012Q3–2019Q1	2014Q1–2019Q1

Notes: i and t index firms and time (survey waves). $F_t^i \pi^{(h)}$ is horizon h -ahead inflation expectation of firm i in wave t . T_t^i is equal to the most recent inflation rate presented to a firm in wave t and 0 for control firms. Seasonal dummies for each sector are included but not reported. The specification is given by equation (1). Standard errors reported in parentheses are as in Driscoll and Kraay (1998). ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(a) Table I: Effect of treatment with past inflation on inflation expectations.

(b) Table II: Duration of effects of signals on inflation expectations.

5.4 Table III: Main effects on prices, employment, credit utilization, and loan applications

Read:

- Panel A shows small and transitory price effects. A 1 percentage point increase in inflation expectations raises reported price changes by about 0.205 percentage points on impact and 0.185 after one quarter, but the effect fades quickly.
- Panels B and C show large negative employment effects. Survey employment and administrative INPS employment both fall persistently after inflation expectations rise.
- Panel D shows positive effects on credit-line utilization. Firms draw more on existing credit lines after inflation expectations rise.
- Panel E shows that loan applications to new financial institutions increase, especially at intermediate horizons.

Economic message: Outside the ELB, higher inflation expectations look stagflationary from the firm's perspective. Firms raise prices only modestly but cut employment and seek more credit, consistent with fears of weaker demand and tighter future financing.

TABLE III
EFFECTS OF INFLATION EXPECTATIONS ON FIRM DECISIONS

	y_t^i (1)	y_{t+1}^i (2)	y_{t+2}^i (3)	y_{t+3}^i (4)	y_{t+4}^i (5)	y_{t+5}^i (6)
Panel A: Effect on prices (source: SIGE)						
$F_{t-1}^i \pi^{(12m)}$	0.205** (0.096)	0.185* (0.097)	0.063 (0.126)	0.008 (0.103)	0.020 (0.092)	-0.077 (0.090)
Observations	16,227	13,765	12,878	12,011	11,208	10,419
R-squared	0.162	0.152	0.129	0.105	0.102	0.097
First-stage F-stat	112.9	113.7	117.2	112.7	112.6	112.5
Panel B: Effect on employment (source: SIGE)						
$F_{t-1}^i \pi^{(12m)}$	-0.074 (0.067)	-0.277* (0.149)	-0.489*** (0.156)	-0.712*** (0.196)	-0.755*** (0.219)	-1.037*** (0.196)
Observations	16,227	13,765	12,878	12,011	11,208	10,419
R-squared	0.018	0.025	0.036	0.055	0.052	0.049
First-stage F-stat	112.9	113.7	117.2	112.7	112.6	112.5
Panel C: Effect on employment (source: INPS)						
$F_{t-1}^i \pi^{(12m)}$	-0.186*** (0.066)	-0.232* (0.131)	-0.609*** (0.209)	-1.081*** (0.285)	-1.195*** (0.283)	-1.522*** (0.447)
Observations	15,062	12,764	11,907	11,090	10,321	9,570
R-squared	0.024	0.032	0.045	0.062	0.063	0.063
First-stage F-stat	114.8	118.4	117.4	114.2	113.5	116.5

(a) Table III, Panels A–C: prices and employment.

TABLE III
CONTINUED

	y_t^i (1)	y_{t+1}^i (2)	y_{t+2}^i (3)	y_{t+3}^i (4)	y_{t+4}^i (5)	y_{t+5}^i (6)
Panel D: Effect on credit utilization (source: CCR)						
$F_{t-1}^i \pi^{(12m)}$	0.234 (0.224)	0.864*** (0.242)	0.730** (0.344)	0.982** (0.467)	1.128** (0.457)	0.964 (0.579)
Observations	13,336	11,261	10,501	9,773	9,085	8,409
R-squared	0.002	0.001	0.001	0.001	-0.001	-0.001
First-stage F-stat	102.9	106.1	104.7	103.3	104.6	107.8
Panel E: Effect on loan applications (source: CCR)						
$F_{t-1}^i \pi^{(12m)}$	0.028* (0.016)	0.031 (0.023)	0.082*** (0.028)	0.072* (0.039)	0.056 (0.051)	0.051 (0.065)
Observations	14,361	12,173	10,501	9,153	8,012	7,037
R-squared	0.007	0.009	0.011	0.015	0.016	0.018
First-stage F-stat	114.8	114.6	117	115.6	115.8	116.8

Notes: i and t index firms and time (survey waves). $F_{t-1}^i \pi^{(12m)}$ is the one-year-ahead inflation expectation of firm i in wave $t-1$. In Panel A, the dependent variable is $y_{t+k}^i = \Delta p_{i,t+k}$ where $\Delta p_{i,t+k}$ is the average change in firm i 's prices over the previous 12 months in period $t+k$. In Panels B and C, the dependent variable is $y_{t+k}^i = \log\left(\frac{L_{i,t+k}}{L_{i,t}} where $L_{i,t}$ is the number of employees in firm i at time t . In Panel D, the dependent variable is $y_{t+k}^i = u_{i,t+k} - u_{i,t-1}$ where $u_{i,t}$ is the utilization rate of credit lines by firm i at time t . In Panel E, the dependent variable is $y_{t+k}^i = \text{App}_{i,t+k} + \dots + \text{App}_{i,t}$ where $\text{App}_{i,t}$ is the number of loan applications made by firm i at time t . The specification is given by equation (2). Seasonal dummies for each sector are included but not reported. Other controls are included but not reported. The estimation sample is 2012Q3–2019Q1. Standard errors reported in parentheses are as in Driscoll and Kraay (1998). ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.$

(b) Table III, Panels D–E: credit utilization and loan applications.

5.5 Table IV: Annual outcomes and balance-sheet adjustment

Read:

- Annual data confirm the employment effect: both SIGE and INPS employment fall when inflation expectations rise.
- Fixed assets fall by about 1.335 percent and sales fall by about 1.850 percent after a 1 percentage point increase in expected inflation.
- Liquidity falls slightly; trade credit falls weakly; cash does not respond significantly.
- Inventories increase significantly, consistent with firms expecting higher raw material prices.
- Leverage rises by about 0.844 percentage points, consistent with the credit-utilization results.
- Wages do not respond significantly, but the employment composition shifts toward white-collar workers and away from temporary workers.

Economic message: Higher inflation expectations change real decisions and balance sheets. Firms shrink activity, hold more inventories, take on more leverage, and reduce flexible labor inputs.

TABLE IV
EFFECTS OF INFLATION EXPECTATIONS ON FIRM DECISIONS, ANNUAL DATA

Row	Outcome variable	Coef. on $F_{t-1}^{(12m)}$ (std. err.) (1)	Obs. (2)	R^2 (3)	First-stage F -stat (4)
(1)	Employment (SIGE)	-1.140*** (0.180)	4,581	0.087	52.15
(2)	Employment (INPS)	-0.575*** (0.227)	4,413	0.135	51.71
(3)	Fixed assets (CADS)	-1.335*** (0.368)	3,142	0.049	35.52
(4)	Sales (CADS)	-1.850*** (0.563)	3,168	0.095	36.96
(5)	Liquidity (current assets to total assets) (CADS)	-0.079** (0.036)	3,180	0.004	36.13
(6)	Trade credit to total assets (CADS)	-0.197* (0.104)	3,180	0.006	36.13
(7)	Cash & liquid financial funds to total assets (CADS)	-0.018 (0.126)	3,180	0.011	36.13
(8)	Inventory to total assets (CADS)	0.224*** (0.027)	3,180	0.007	36.13

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TABLE IV
CONTINUED

Row	Outcome variable	Coef. on $F_{t-1}^{(12m)}$ (std. err.) (1)	Obs. (2)	R^2 (3)	First-stage F -stat (4)
(9)	Leverage ratio (CADS)	0.844*** (0.258)	3,180	0.007	36.13
(10)	Average wages (INPS)	0.466 (0.432)	4,413	0.011	50.88
(11)	Average wages of white-collar workers (INPS)	0.277 (0.273)	4,375	0.007	53.07
(12)	Average wages of blue-collar workers (INPS)	-0.093 (0.683)	3,975	0.012	51.95
(13)	Employment share of white-collar workers (INPS)	0.271*** (0.035)	4,413	0.009	50.88
(14)	Employment share of temporary workers (INPS)	-0.589*** (0.147)	4,414	0.013	50.76

Notes: The table reports coefficient α_0 in specification (3) estimated at the annual frequency. The second column indicates the outcome variable (the source of each variable is indicated in parentheses). The dependent variable is $\ln\left(\frac{Y_{i,t+h}}{Y_{i,t-1}}\right) \times 100$ where $Y_{i,t}$ is the value of an outcome variable for firm i in year t . Ratios and shares are measured in percent and the corresponding dependent variable is $Y_{i,t+h} - Y_{i,t-1}$. Horizon h is set at one. [Driscoll and Kraay \(1998\)](#) standard errors are reported in parentheses. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(a) Table IV, rows 1–8: annual real and asset-side outcomes.

(b) Table IV, rows 9–14: leverage, wages, and labor composition.

5.6 Table V: Heterogeneity in the effects of inflation expectations

Read:

- Panel A splits firms by sector. Construction firms show stronger price and employment responses than manufacturing and services.
- Panel B splits by employment size. Credit utilization responses are positive across size groups, while employment reductions appear across groups with some variation.
- Panel C splits by export share. Low-export firms show stronger employment declines, consistent with domestic demand exposure.
- Panel D splits by geography. Firms in southern Italy respond more strongly, especially through employment reductions and credit utilization.

Economic message: Inflation expectations matter most for firms more exposed to domestic demand, credit conditions, and local economic fragility. The effects are not homogeneous across the Italian economy.

TABLE V
HETEROGENEITY IN EFFECTS OF INFLATION EXPECTATIONS ON PRICES, EMPLOYMENT, AND CREDIT

	Dependent variable: $d p_{i,t+1}$			Dependent variable: $\log \left(\frac{h_{i,t+1}}{h_{i,t}} \right)$			Dependent variable: $u_{i,t+1} - u_{i,t-1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Sector									
	Manufacturing	Services	Construction	Manufacturing	Services	Construction	Manufacturing	Services	Construction
$F_{i,t-1}^{(120)}$	0.145 (0.136)	0.186** (0.080)	0.620*** (0.218)	-0.484 (0.365)	-0.893*** (0.296)	-0.371 (0.842)	-0.477 (0.374)	0.602 (0.493)	0.483 (0.851)
Observations	5,648	5,665	2,452	5,648	5,665	2,452	5,351	5,209	2,224
R-squared	0.009	0.157	0.215	0.016	0.009	0.016	0.004	0.011	0.004
First-stage F-stat	105	123.3	51.86	105	123.3	51.86	107.9	130.9	48.57
Panel B: Number of employees									
	50-99	100-299	300 or more	50-99	100-299	300 or more	50-99	100-299	300 or more
$F_{i,t-1}^{(120)}$	0.160 (0.151)	0.149*** (0.068)	0.260*** (0.112)	-0.708*** (0.340)	-0.366*** (0.464)	-0.735*** (0.428)	1.540*** (0.427)	0.709*** (0.410)	1.085*** (0.430)
Observations	4,767	4,009	4,989	4,767	4,009	4,989	3,895	3,523	3,736
R-squared	0.160	0.131	0.198	0.003	0.025	0.018	0.003	0.003	0.001
First-stage F-stat	94.68	134.5	119.9	94.68	134.5	119.9	93.29	120	101.1

(a) Table V, Panels A–B: heterogeneity by sector and size.

TABLE V
CONTINUED

	Dependent variable: $d p_{i,t+1}$			Dependent variable: $\log \left(\frac{h_{i,t+1}}{h_{i,t}} \right)$			Dependent variable: $u_{i,t+1} - u_{i,t-1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C: Export share, percent									
	0	1-33	34 or more	0	1-33	34 or more	0	1-33	34 or more
$F_{i,t-1}^{(120)}$	0.137 (0.093)	0.173 (0.109)	0.248 (0.167)	-1.121*** (0.428)	-0.583 (0.414)	-0.093 (0.277)	0.962*** (0.324)	1.235*** (0.496)	0.551 (0.422)
Observations	6,360	2,905	4,500	6,360	2,905	4,500	5,048	2,399	3,814
R-squared	0.168	0.184	0.113	0.005	0.029	0.012	0.002	0.011	0.001
First-stage F-stat	115.3	109.1	100	115.3	109.1	100	99.74	104.9	97.77
Panel D: Geography									
	North	Center	South	North	Center	South	North	Center	South
$F_{i,t-1}^{(120)}$	0.174* (0.097)	0.078 (0.187)	0.307*** (0.105)	-0.635* (0.332)	-0.039 (0.439)	-1.156*** (0.548)	0.903*** (0.359)	0.079 (0.617)	1.702*** (0.459)
Observations	8,007	2,309	2,819	8,007	2,309	2,819	6,584	2,426	2,549
R-squared	0.155	0.191	0.107	0.014	0.018	0.015	-0.001	0.012	0.004
First-stage F-stat	109.3	106.7	108	109.3	106.7	108	106.1	77.79	112

Notes: i and t index firms and time (survey waves). $F_{i,t-1}^{(120)}$ is the one-year-ahead inflation expectation of firm i in wave t . Treatment is equal to the most recent inflation rate presented to a firm for treated firms and 0 for control firms. Seasonal dummies for each sector are included but not reported. Other controls from Table III are included but not reported. The estimation sample is 2013Q3–2019Q1. Standard errors reported in parentheses are as in [Driscoll and Kraay \(1998\)](#). ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(b) Table V, Panels C–D: heterogeneity by exports and geography.

5.7 Table VI: Other expectations and plans

Read:

- Higher inflation expectations lead firms to report worse aggregate economic conditions and a lower probability of improvement.
- Firms also expect worse own-business conditions, lower product demand, worse liquidity, and worse credit access.
- Uncertainty rises at both short and medium horizons.
- Firms plan lower employment and lower investment, consistent with the actual employment and fixed-asset responses.
- Expected price changes rise weakly, but demand is expected to push prices down while raw material prices are expected to push prices up.

Economic message: The mechanism is not a simple New Keynesian demand channel in the full sample. Italian firms interpret higher inflation as bad news about aggregate and firm-specific conditions, with cost pressure from raw materials but weak demand.

TABLE VI
EFFECTS OF INFLATION EXPECTATIONS ON OTHER EXPECTATIONS AND PLANS

Row	Outcome variable	Coef. on $F_{i,t-1}^{(120)}$ (std. err.)	Obs. (1)	R^2 (2)	First-stage F-stat (4)
Macroeconomic conditions					
(1)	General economic situation relative to 3 months ago	-0.235*** (0.039)	20,256	-0.021	144.30
(2)	Probability of improved situation in the next 3 months	-2.287*** (0.553)	20,432	-0.009	145.16
Firm-specific conditions					
(3)	Expected business conditions for company, next 3 months	-0.160*** (0.023)	20,421	-0.003	146.28
(4)	Expected demand for products, next 3 months	-0.104*** (0.026)	19,033	-0.008	95.45
(5)	Expected liquidity for company, next 3 months	-0.073*** (0.018)	20,181	0.000	146.33
(6)	Access condition to credit relative to 3 months ago	-0.118*** (0.012)	20,115	0.001	146.1
(7)	Uncertainty 3-month ahead	0.005** (0.003)	20,110	0.001	145.50
(8)	3-year ahead	0.009*** (0.002)	20,122	0.001	147.84

(a) Table VI, rows 1–8: macro, firm-specific, and uncertainty expectations.

TABLE VI
CONTINUED

Row	Outcome variable	Coef. on $F_{i,t-1}^{(120)}$ (std. err.)	Obs. (1)	R^2 (2)	First-stage F-stat (4)
(9)	Expected employment change, next 3 months	-0.066*** (0.014)	20,379	0.002	144.29
(10)	Expected investment change, next calendar year	-0.117*** (0.041)	18,282	-0.004	114.17
(11)	Expected price change, next 12 months	0.100* (0.057)	20,512	0.003	146.31
(12)	Factors affecting future price changes	-0.134*** (0.020)	19,956	-0.002	147.08
(13)	Expected raw material prices	0.083*** (0.026)	19,894	0.003	147.99
(14)	Expected labor costs	0.018 (0.013)	19,912	0.001	147.03
(15)	Expected prices of competitors	-0.033 (0.022)	19,870	0.001	147.75

Notes: i and t index firms and time (survey waves). The specification is given by equation (4). $F_{i,t-1}^{(120)}$ is the one-year-ahead inflation expectation of firm i in wave $t-1$. The second column reports the dependent variables. $F_{i,t-1}^{(120)}$ is instrumented with the treatment variable. Seasonal dummies for each sector are included but not reported. The estimation sample is 2013Q3–2019Q1. Standard errors reported in parentheses are as in [Driscoll and Kraay \(1998\)](#). ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(b) Table VI, rows 9–15: plans and price-change factors.

5.8 Tables VII and VIII: Profits and ELB firm decisions

Read:

- Table VII shows that treated firms earn higher profit shares during the treatment period. The estimated effect ranges from 0.172 to 0.284 percentage points depending on controls.
- There is no corresponding profit difference in the pretreatment period. This supports the interpretation that the information treatment modestly improves firm decisions.
- Table VIII focuses on the ELB period. During the ELB, higher inflation expectations produce larger and more persistent price increases than in the full sample.
- Employment effects become statistically insignificant during the ELB, and point estimates are much less negative.
- Credit utilization responses become larger by longer horizons.

Economic message: The profit result supports rational inattention: information is useful but only modestly profitable. The ELB result supports the idea that inflation expectations become more expansionary when nominal rates are constrained.

TABLE VII
PROFITS OF TREATMENT AND CONTROL GROUP FIRMS

Dependent variable: profit share in sales	Controls (1)	Year and industry fixed effects (2)	No control (3)
Panel A: Treatment period, 2012Q3–2019Q1			
$Treatment_{it-1}^i$	0.284*** (0.076)	0.172** (0.085)	0.179** (0.080)
Observations	4,077	4,046	4,033
R-squared	0.120	0.001	0.001
Panel B: Pretreatment period, 2006Q1–2012Q2			
$Treatment_{2012Q4}^i$	−0.023 (0.045)	−0.065 (0.065)	−0.027 (0.065)
Observations	2,032	2,026	2,036
R-squared	0.063	0.000	0.000

Notes. The table reports results using Huber-robust regressions. The dependent variable is the profit share in sales (profits/sales $\times$ 100). $Treatment_{it}^i$ is a dummy variable equal to 1 if a firm i in year t is provided information about past inflation and 0 otherwise. Column (1) includes controls as in Table III. Column includes only industry fixed effects and year fixed effects. Column (3) has no controls. Panel A reports re

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(a) Table VII: Profits of treatment and control group firms.

(b) Table VIII: ELB-period effects on firm decisions.

5.9 Table IX: ELB-period expectations and plans

Read:

- During the ELB, firms with higher inflation expectations become more optimistic rather than pessimistic about macroeconomic conditions.
- Expected demand, expected liquidity, planned employment, planned investment, and expected price changes all move in a more expansionary direction.
- Firms report that demand, raw material prices, and competitor prices all push expected prices upward during the ELB.

- The contrast with Table VI is central: the interpretation of inflation news changes sharply across regimes.

Economic message: At the ELB, higher inflation expectations behave more like a demand stimulus. This is consistent with the New Keynesian logic that higher expected inflation lowers real interest rates when nominal rates do not adjust upward.

TABLE IX
THE ELB PERIOD: EFFECTS OF INFLATION EXPECTATIONS ON OTHER EXPECTATIONS AND PLANS

Row	Outcome variable	Coef. on $\pi_{t-1}^{(12m)}$ (std. err.) (1)	Obs. (2)	R^2 (3)	First-stage F -stat (4)
Macroeconomic conditions					
(1)	General economic situation relative to 3 months ago	0.116 (0.110)	11,991	-0.007	91.05
(2)	Probability of improved situation in the next 3 months	2.211 (1.677)	12,136	0.010	88.23
Firm-specific conditions					
(3)	Expected business conditions for company, next 3 months	0.076 (0.047)	12,125	-0.006	88.60
(4)	Expected demand for products, next 3 months	0.039* (0.022)	11,971	0.000	85.23
(5)	Expected liquidity for company, next 3 months	0.100*** (0.040)	11,989	-0.017	88.95
(6)	Access condition to credit relative to 3 months ago	-0.004 (0.023)	11,918	-0.000	91.74
(7)	Uncertainty 3-month ahead	0.030*** (0.006)	11,902	0.002	87.52
(8)	3-year ahead	0.002 (0.005)	11,918	0.001	88.58

(a) Table IX, rows 1–8: ELB-period macro and firm expectations.

TABLE IX
CONTINUED

Row	Outcome variable	Coef. on $\pi_{t-1}^{(12m)}$ (std. err.) (1)	Obs. (2)	R^2 (3)	First-stage F -stat (4)
(9)	Expected employment change, next 3 months	0.091*** (0.026)	12,111	-0.011	88.78
(10)	Expected investment change, next calendar year	0.119*** (0.038)	12,008	-0.002	90.16
(11)	Expected price change, next 12 months	0.384*** (0.096)	12,177	0.012	89.75
Factors affecting future price changes					
(12)	Expected change in demand	0.102** (0.048)	11,819	0.000	87.46
(13)	Expected raw material prices	0.275*** (0.066)	11,790	-0.026	87.78
(14)	Expected labor costs	-0.003 (0.046)	11,798	-0.000	85.78
(15)	Expected prices of competitors	0.171*** (0.038)	11,760	-0.009	89.54

Notes. i and t index firms and time (survey waves). The specification is given by equation (4). $\pi_{t-1}^{(12m)}$ is the one-year-ahead inflation expectation of firm i in wave $t-1$. The second column indicates the dependent variables. $\pi_{t-1}^{(12m)}$ is instrumented with the treatment variable lagged one quarter. Seasonal dummies for each sector are included but not reported. The estimation sample is 2011Q4–2018Q4. Standard errors reported in parentheses are as in Deaton and Cartwright (1998). ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(b) Table IX, rows 9–15: ELB-period plans and price factors.

6 Short summary

- The paper uses randomized information treatment in a Bank of Italy firm survey to generate exogenous variation in inflation expectations.
- Treated firms receive recent inflation information; control firms do not.
- The treatment strongly moves inflation expectations, especially at short horizons.
- The treatment effect on expectations fades quickly, suggesting that persistent communication is necessary for persistent belief changes.
- In the full sample, higher inflation expectations lead firms to raise prices slightly, reduce employment, reduce fixed assets, reduce sales, increase credit utilization, raise inventories, and raise leverage.
- Firms interpret higher inflation as bad news outside the ELB: worse aggregate conditions, worse business conditions, lower demand, and worse credit access.
- Treated firms earn modestly higher profit shares, consistent with rational inattention.
- During the ELB, higher inflation expectations produce stronger price increases and no employment decline.
- During the ELB, firms interpret higher inflation more positively: better demand, better liquidity, and more planned employment and investment.
- The main lesson is that expectations causally matter for firm decisions, but the sign and mechanism depend on the macroeconomic regime.

7 Conclusion

7.1 Core conclusion

Coibion, Gorodnichenko, and Ropele provide causal evidence that firms' inflation expectations affect firm decisions. By exploiting a randomized information treatment, they show that inflation expectations are not passive survey responses. They influence prices, employment, credit demand, investment, balance sheets, and profits.

7.2 Economic intuition

The effect of inflation expectations depends on what firms infer from inflation. In the full Italian sample, higher inflation is interpreted as a negative signal: it predicts higher raw material prices, weaker demand, worse liquidity, and worse credit access. Firms respond by cutting employment and fixed assets while raising prices only modestly. During the ELB, however, higher inflation is interpreted as more demand-oriented or expansionary, because higher expected inflation lowers real interest rates when nominal policy rates are constrained.

7.3 Incremental contribution revisited

The paper's contribution is the combination of randomized expectations variation, long panel survey data, rich administrative outcome data, and an explicit comparison between normal times and the ELB. This allows the authors to make a stronger causal claim than studies relying on correlations between expectations and decisions.

7.4 Policy implications

The results support the relevance of central bank communication. If policymakers can reach firms directly and repeatedly, they can change expectations and thereby change firm behavior. However, the findings also warn that the interpretation of inflation communication is context dependent. The same inflation information may be contractionary in one environment and expansionary in another.

7.5 Limitations

The treatment provides recent inflation information rather than a full central bank communication package. The treated firms are a small subset of the economy, so the estimated effects do not include general-equilibrium feedbacks that would arise if all firms changed expectations simultaneously. The paper also cannot fully distinguish whether the ELB results reflect the mechanical real-rate channel or a change in the composition of underlying shocks during the ELB.

7.6 Takeaway

The enduring takeaway is that expectations policy can work through firms, not only households and financial markets. But expectations management is not just about moving a forecast number.

It also depends on how firms interpret the information, what they infer about demand and costs, and whether monetary policy is constrained by the effective lower bound.